

Supplementary Table 1. Social breeding system and kin structure of breeding group for cooperatively breeding birds. Classifications are as follows:

Column 2: Primary pattern of dispersal that gives rise to cooperative groups. D = delayed dispersal (offspring remain on natal territory after fledging); I = immigration (unrelated individual joins existing breeding group); C = coalition (several same-sex individuals form a group to establish a breeding site or territory).

Column 3: Frequency of cooperative breeding by non-kin, given as the percentage of helpers in the population that are unrelated to the breeding pair (for helper-at-the-nest systems); or as the percentage of adult group members in the population who are unrelated to any other group member (for cooperatively polygamous species). “Majority” indicates that non-kin cooperation is the typical breeding system; “frequent” indicates that exact percentages were not given but that the available data suggests that non-kin cooperation is common in the population; a long dash (—) indicates that kin-based cooperation is the typical breeding system and that unrelated immigrants account for fewer than 20% of adult group members or helpers.

Column 4: Type of data used to calculate kin relationships in the social group. B = behavioural observations (for kin groups only, including observations of recently fledged juveniles feeding apparent younger siblings; or when authors suspected helpers to be non-dispersed offspring without evidence from marked individuals); R = colour-ringing data, based on known genealogy of marked individuals ; G = genetic data, from microsatellite genotyping, DNA fingerprinting, or protein electrophoresis.

Column 5: Frequency of cooperative breeding in the population. O = obligate (>95% of breeding units are cooperative groups); F = facultative (>5% of breeding units are lone pairs). “Obligate” is used to indicate that nesting by lone pairs was rare in the population studied, not that the species has necessarily lost solitary breeding entirely.

Column 6: Kin structure of social group: primarily composed of kin (K); composed of non-kin (NK); or mixed (M, regularly contain both related and unrelated helpers or co-breeding adult group members).

Column 7: Social breeding system, when cooperative; genetic mating patterns are also given when it is known that they deviate from the social mating pattern.

Column 8: Classification of social breeding system: 1 = pair-nesting with related helpers; 2 = pair-nesting with unrelated helpers or with both related and unrelated helpers; 3 = cooperative polygamy or joint-nesting by non-relatives; 4 = cooperative polygamy or joint-nesting by relatives and non-relatives, with related or unrelated helpers.

Higher Taxon 1. Scientific name	2. Dispersal	3. Frequency	4. Data	5. O/F	6. Kin	7. Social breeding system (when cooperative)	8. Category	9. Source
Ratitae: Rheidae								
<i>Rhea americana</i>	C	majority	R	F	NK	cooperative polyandry by unrelated males; male-only parental care with subordinate male helper (kinship unknown)	3	[1, 2]
Ratitae: Apterygidae								
<i>Apteryx australis</i>	D	—	B	F	K	pair nesting with (presumed) related male helpers	1	[3]
Galliformes: Phasianidae								
<i>Tetraophasis szechenyii</i>	D	—	R	F	K	pair nesting with (presumed) related male helpers	1	[4, 5]
Anseriformes: Anseratinae								
<i>Anseranas semipalmata</i>	C	majority	G	F	NK	cooperative polygyny (joint nesting) by unrelated females	3	[6, 7]
Piciformes: Ramphastidae								
<i>Semnornis ramphastinus</i>	D	—	R	F	K	pair nesting with related male and female helpers	1	[8]
Piciformes: Picidae								
<i>Melanerpes formicivorus</i>								
Pop'n 1 (Hastings, CA)	D	—	R	F	K	cooperative polygynandry (joint nesting) by related breeders with related male and female helpers	4	[9, 10]
Pop'n 2 (Water Canyon, NM)	D, C	25-50%	R	F	M	cooperative polygynandry (joint nesting) by related and unrelated breeders with related male and female helpers	4	[11]
<i>Picoides borealis</i>	D	—	G	F	K	pair nesting with related male helpers	1	[12]
Coraciiformes: Bucorvidae								
<i>Bucorvus abyssinicus</i>	D	—	B	F	K	pair nesting with (presumed) related male helpers	1	[13]
<i>Bucorvus leadbeateri</i>	D	—	B	F	K	pair nesting with (presumed) related male helpers	1	[14, 15]
<i>Anorrhinus austeni</i>	D	—	B	F	K	pair nesting with (presumed) related male and female helpers	1	[16]
<i>Anorrhinus tickelli</i>	D	—	B	F	K	pair nesting with (presumed) related male and female helpers	1	[16]
<i>Anorrhinus galeritus</i>	D	—	B	F	K	pair nesting with (presumed) related male and female helpers	1	[16]
Coraciiformes: Upupidae								
<i>Upupa epops</i>	I	majority	G	F	NK	pair nesting with unrelated male helpers; possible cooperative polyandry	2	[17]

Coraciiformes: Phoeniculidae								
<i>Phoeniculus purpureus</i>	D, I	25-30%	R	F	M	pair nesting with related and unrelated male and female helpers	2	[18-20]
<i>Phoeniculus damarensis</i>	D, I	25-30%	B	F	M	pair nesting with related and unrelated male and female helpers	2	[21]
Coraciiformes: Meropidae								
<i>Merops bulocki</i>	D	—	B	F	K	pair nesting with (presumed) related male helpers ; widespread conspecific brood parasitism	1	[22]
<i>Merops bullockoides</i>	D	—	R	F	K	pair nesting with related male and female helpers; widespread conspecific brood parasitism	1	[23, 24]
<i>Merops ornatus</i>	D	—	B	F	K	pair nesting with (presumed) related male helpers	1	[25]
<i>Merops apiaster</i>	D	—	G	F	K	pair nesting with related male helpers	1	[26]
<i>Merops orientalis</i>	D	—	B	F	K	pair nesting with (presumed) related helper	1	[27]
Coraciiformes: Cerylidae								
<i>Ceryle rudis</i>	D, I	50%	R	F	M	pair nesting with related and unrelated male helpers	2	[28, 29]
Coraciiformes: Dacelonidae								
<i>Dacelo novaeguinea</i>	D	—	G	F	K	pair nesting with related male and female helpers	1	[30, 31]
<i>Dacelo leachii</i>	D	—	R	F	K	pair nesting with related male helpers	1	[32]
<i>Todiramphus cinnamominus</i>	D	—	R	F	K	pair nesting with related male and female helpers	1	[33]
<i>Tanysiptera sylvia</i>	I	majority	R	F	NK	pair nesting with unrelated male helpers	2	[34]
Coraciiformes: Todidae								
<i>Todus mexicanus</i>	I	majority	R	F	NK	pair nesting with unrelated male helpers	2	[35]
Coraciiformes: Coliidae								
<i>Colius striatus</i>	D, I	frequent ¹	R	O	M	cooperative polygyny (joint nesting) by unrelated females and related and unrelated male helpers	4	[36, 37]
<i>Urocolius indicus</i>	D, I	majority	R	F	NK	cooperative polygyny (joint nesting) by unrelated females	3	[37]

¹ Decoux [36] does not give the percentage of unrelated group members, but notes that juvenile dispersal and adult mortality are high and that changes in group membership are frequent; he concludes that genetic relatedness in groups is too low for the social system to be favored by kin selection.

Cuculiformes: Cuculidae								
<i>Guira guira</i>	D, C, I	frequent ²	G	O	M	cooperative polygynandry and joint nesting by related and unrelated adults; related male helpers	4	[38]
<i>Crotophaga major</i>	D, C, I	majority	G	O	NK	joint nesting by unrelated pairs; related male helpers (uncommon); conspecific brood parasitism	3	[39, 40]
<i>Crotophaga ani</i>	D, C, I	majority	R	F	NK	joint nesting by unrelated pairs; related male helpers (uncommon)	3	[41]
<i>Crotophaga sulcirostris</i>	D, C, I	majority	R	F	NK	joint nesting by unrelated pairs; related male helpers (uncommon)	3	[42, 43]
Opisthocomidae								
<i>Opisthocomus hoatzin</i>	D, I	15-20%	G	F	M	pair nesting with related male helpers ; joint nesting by unrelated pairs	4	[44]
Psittaciformes: Psittacidae								
<i>Electus roratus</i>	C	majority	G	F	NK	cooperative polyandry by unrelated males	3	[45, 46]
<i>Guarouba guarouba</i>	D	—	G	F	K	pair nesting with related helpers	1	[47, 48]
<i>Caracopsis vasa</i>	I	majority	G	O	NK	cooperative polyandry by unrelated males	3	[49]
<i>Pyrrhura orcesi</i>	D	—	G	F	K	pair nesting with related male and female helpers	1	[50]
<i>Myiopsitta monachus</i>	D	—	B	F	K	pair nesting with (presumed) related male and female helpers	1	[51]
<i>Cyanoramphus saisseti</i>	I	majority	G	O	NK	cooperative polyandry by unrelated males	3	[52]
Trochiliformes: Apodidae								
<i>Collocalia terraereginae</i>	D	—	R	F	K	pair nesting with related male and female helpers	1	[53, 54]
<i>Chaetura pelagica</i>	D, I	30-60%	R	F	M	pair nesting with related and unrelated male and female helpers	2	[55]
Musophagiformes: Musophagidae								
<i>Corythaeola cristata</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[56]
<i>Corythaixoides concolor</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[56]
<i>Corathaixoides leucogaster</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[56]
Gruiformes: Psophiidae								
<i>Psophia crepitans</i>	D, C, I	majority	B	O	M	cooperative polyandry by unrelated males; related and unrelated male and female helpers	4	[57]

² Mean relatedness among males in groups was significantly higher than expected by chance for 6 of 20 groups (30%); this included non-breeding helpers as well as breeding adults. Data were not available for relatedness among females [38].

<i>Psophia leucoptera</i>	D, C, I	majority	R	O	M	cooperative polyandry by unrelated males; related and unrelated male and female helpers	4	[58, 59]
Gruiformes: Rallidae								
<i>Gallirallus sylvestris</i>	D	—	B	F	K	pair nesting with related helpers	1	[60]
<i>Amaurornis flavirostra</i>	D	—	B	F	K	pair nesting with related helpers	1	[60]
<i>Porzana atra</i>	D	—	B	F	K	pair nesting with related helpers	1	[60]
<i>Porphyrio porphyrio</i>								
Pop'n 1 (NZ South)	D	—	G	O	K	cooperative polygynandry by related males and females	4	[61-63]
Pop'n 2 (NZ North)	C, I	majority	G	F	NK	cooperative polygynandry by unrelated males and females	3	[61-63]
<i>Porphyrio mantelli</i>	D	—	B	F	K	pair nesting with related helpers	1	[60]
<i>Porphyrio martinicus</i>	D	—	R	F	K	pair nesting and cooperative polygyny by related females with related helpers	4	[64]
<i>Gallinula nesiotis</i>	D	—	B	F	K	pair nesting with related helpers	1	[60]
<i>Gallinula chloropus</i>	D	—	R	F	M	pair nesting and cooperative polygyny by related females with related helpers; conspecific brood parasitism	4	[65, 66]
<i>Gallinula tenebrosa</i>	D, C, I	majority	G	O	M	cooperative polygynandry by unrelated males and females with related helpers	4	[67]
<i>Gallinula mortierii</i>	D, I	25-30%	R	F	M	cooperative polygynandry by related females and related and unrelated males, with related helpers	4	[68-70]
<i>Fulica cristata</i>	D	—	B	F	K	pair nesting with related helpers (offspring from earlier brood of year)	1	[60]
<i>Fulica atra</i>	D	—	B	F	K	pair nesting with related helpers (offspring from earlier brood of year)	1	[60, 71]
<i>Fulica ardesiaca</i>	D	—	B	F	K	pair nesting with related helpers (offspring from earlier brood of year)	1	[60]
<i>Fulica gigantea</i>	D	—	B	F	K	pair nesting with related helpers (offspring from earlier brood of year)	1	[60]
Gruiformes: Rhynochetidae								
<i>Rhynochetos jubatus</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[72]
Gruiformes: Mesitornithidae								
<i>Monias benschi</i>	D, I	frequent ³	G	F	M	cooperative polygynandry with related and unrelated male and female helpers	4	[73]
Charadriiformes: Stercorariidae								

³ Of 9 groups with multiple offspring, 6 were composed of a breeding pair with unrelated helpers, and 3 were composed of multiple male and female breeders with related helpers. Of 14 completely sampled groups, 5 contained adults that were more closely related to each other than chance would predict, and 11 did not [73].

<i>Catharacta lonnbergi</i>	C	majority	R	F	NK	cooperative polyandry by unrelated males	3	[74, 75]
Charadriiformes: Charadriidae								
<i>Vanellus chilensis</i>	I	majority	G	F	NK	cooperative polyandry by unrelated males	3	[76]
Charadriiformes: Haematopodidae								
<i>Haematopus ostralegus</i>	C	majority	G	F	NK	cooperative polygyny by unrelated females	3	[77]
Procellariiformes: Diomedidae								
<i>Phoebastria immutabilis</i>	C	majority	G	F	NK	cooperative polygyny by unrelated females	3	[78]
Falconiformes: Accipitridae								
<i>Ictinia mississippiensis</i>	D	—	B	F	K	pair nesting with (presumed) related male and female helpers	1	[79]
<i>Haliaeetus vociferoides</i>	C, I	majority	G	F	NK	cooperative polyandry by unrelated males	3	[80]
<i>Gypaetus barbatus</i>	C, I	majority	B	F	NK	cooperative polyandry by (presumed) unrelated males	3	[81]
<i>Melierax canorus</i>	C, I	majority	R	F	NK	cooperative polyandry by (presumed) unrelated males	3	[82]
<i>Parabuteo unicinctus</i>	D, C, I	0-68%	R	F	M	cooperative polyandry by unrelated males with related and unrelated male and female helpers	4	[83, 84]
<i>Buteo galapagoensis</i>	C	majority	G	F	NK	cooperative polyandry by unrelated males	3	[85, 86]
Falconiformes: Falconidae								
<i>Falco columbarius</i>	I	majority	R	F	NK	pair nesting with unrelated male helper	2	[87, 88]
Acanthisittidae								
<i>Acanthisitta chloris</i>	D, I	majority	R	F	M	pair nesting with related and unrelated male helpers	2	[89, 90]
Furnariidae								
<i>Phacellodomus rufifrons</i>	D	—	R	F	K	pair nesting with related male helpers	1	[91]
<i>Anumbius annumbi</i>	D	—	R	F	K	pair nesting with related helpers	1	[92]
<i>Coryphistera alaudina</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[93]
<i>Acrobatornis fonsecai</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[94]
<i>Pseudoseisura lophotes</i>	D	—	B	F	K	pair nesting with related helpers	1	[95]
Climacteridae								
<i>Climacteris affinis</i>	D	—	R	F	K	pair nesting with related male helpers	1	[96]
<i>Climacteris erythrops</i>	D	—	R	F	K	pair nesting with related male helpers	1	[97]

<i>Climacteris picumnus</i>	D	—	R	F	K	pair nesting with related male helpers	1	[98]
<i>Climacteris melanura</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[99]
<i>Climacteris rufa</i>	D	—	R	F	K	pair nesting with related male and female helpers	1	[100]
Maluridae								
<i>Malurus alboscapulatus</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[101]
<i>Malurus melanocephalus</i>	D	frequent ⁴	G	F	M	pair nesting with related and unrelated male helpers; polyandrous mating with extra-group males	2	[102-104]
<i>Malurus leucopterus</i>	D	frequent ⁴	R	F	M	pair nesting with related and unrelated male helpers; polyandrous mating with extra-group males	2	[105, 106]
<i>Malurus cyaneus</i>	D	39% ⁵	G	F	M	pair nesting with related and unrelated male helpers; polyandrous mating with extra-group males	2	[107]
<i>Malurus splendens</i>	D	frequent ⁴	G	F	M	pair nesting with related and unrelated male helpers; polyandrous mating with extra-group males	2	[108]
<i>Malurus lamberti</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[101]
<i>Malurus amabilis</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[101]
<i>Malurus elegans</i>	D	frequent ⁴	G	F	M	pair nesting with related and unrelated male and female helpers; polyandrous mating with extra-group males	2	[109, 110]
<i>Malurus pulcherrimus</i>	D	—	R	F	K	pair nesting with (presumed) related male and female helpers	1	[111]
<i>Malurus coronatus</i>	D	—	R	F	K	pair nesting with (presumed) related male and female helpers	1	[112]
<i>Malurus cyanocephalus</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[101]
<i>Stipiturus malachurus</i>	D	—	R	F	K	pair nesting with (presumed) related male helpers	1	[113]
Meliphagidae								
<i>Lichenostomus melanops</i>	D	—	R	F	K	pair nesting with related male and female helpers	1	[114]
<i>Manorina melanophrys</i>	D, I	50%	G	O	M	pair nesting with related and unrelated male and female helpers	2	[115, 116]
<i>Manorina melanocephala</i>	D	—	B	O	K	pair nesting with related male helpers	1	[117]
<i>Notiomystis cincta</i>	C, I	majority	R	F	NK	cooperative polyandry by unrelated males; polygynandrous mating	3	[118]
Acanthizidae								
<i>Sericornis frontalis</i>	D, I	20%	G	F	M	pair nesting with related and unrelated male helpers; cooperative polyandry	2	[119]

⁴ In these species of *Malurus* fairy-wrens, high extra-pair copulation rates result in low genetic relatedness between helpers and the young that they provision.

⁵ Percentage of helpers that were unrelated to any of the young in the brood that they provisioned.

<i>Acanthiza reguloides</i>	D, I	frequent ⁶	R	F	M	pair nesting with related and unrelated male helpers	2	[120]
<i>Acanthiza chrysorrhoa</i>	D	—	R	F	K	pair nesting with related helpers	1	[121]
<i>Acanthiza lineata</i>	D, I	frequent ⁶	R	F	M	pair nesting with related and unrelated male helpers	2	[120]
Pomatostomidae								
<i>Pomatostomus temporalis</i>	D, I	—	G	F	M	pair nesting with related and unrelated male and female helpers	2	[122]
<i>Pomatostomus superciliosus</i>	D	—	R	F	K	pair nesting with related male and female helpers; plural nesting	1	[123]
<i>Pomatostomus ruficeps</i>	D, I	25-32%	G	O	M	pair nesting with related and unrelated male helpers	2	[124-126]
Neosittidae								
<i>Daphoenositta chrysoptera</i>	D, I	frequent ⁶	R	F	M	pair nesting with related male and unrelated female helpers	2	[127]
Campephagidae								
<i>Pteropodocys maxima</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[128]
Artamidae								
<i>Cracticus torquatus</i>	D	—	B	F	K	pair nesting with (presumed) related helper	1	[129]
<i>Cracticus nigrogularis</i>	D, I	frequent ⁷	G	O	M	pair nesting with related and unrelated male and female helpers; polyandrous mating; conspecific brood parasitism	2	[130]
<i>Gymnorhina tibicen</i>	D, I	frequent ⁸	G	O	M	pair nesting with related and unrelated male and female helpers; polygynandrous mating with extra-group males; plural nesting	2	[131]
<i>Artamus cyanopterus</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[132]
<i>Artamus cinereus</i>	D	—	R	F	K	pair nesting with related helpers	1	[133]
Malaconotidae								
<i>Prionops plumatus</i>	D	—	R	O	K	pair nesting with related (sibling) male and female helpers	1	[134]
<i>Schetba rufa</i>	D, I	frequent ⁹	G	F	M	pair nesting with related and unrelated male helpers	2	[135, 136]
<i>Falcula palliata</i>	C	majority	R	F	NK	cooperative polyandry by unrelated males	3	[137]
<i>Oriolia bernieri</i>	C	majority	B	F	NK	cooperative polyandry by unrelated males	3	[138]
Corcoracidae								

⁶ In these species, breeding units amalgamate into second-tier groups called “clans;” failed breeders within clans often join successful pairs to feed young after fledging. In many cases clan members seem likely to be relatives, but in others cooperative groups are members of different clans and are unlikely to be related.

⁷ 37% of nests are parasitized by extra-group females and multiple mating is common, leading to low genetic relatedness between helpers and offspring [130].

⁸ 82% of offspring are sired by extra-group males and 10% are the result of parasitism by extra-group females, so adult males typically care for offspring that are not their own. Although genetic relatedness between helpers and offspring is low, it is higher than expected by chance among breeding adults in the group.

⁹ Related helpers are one-year-old sons; unrelated helpers are older males. Sample sizes not given, but unrelated and related helpers provisioned nests at similar rates.

<i>Corcorax melanorhamphos</i>	D, C, I	frequent ¹⁰	G	O	M	pair nesting with related and unrelated male and female helpers; cooperative polyandry by related and unrelated males	2	[139-142]
<i>Struthidea cinerea</i>	D	—	G	O	K	pair nesting with related male and female helpers	1	[143, 144]
Laniidae								
<i>Lanius excubitoroides</i>	D	—	R	F	K	pair nesting with related male and female helpers	1	[145]
<i>Corvinella corvina</i>	D, I	frequent ¹¹	R	O	M	pair nesting with related and unrelated male and female helpers	2	[146]
<i>Corvinella melanoleuca</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[147]
Corvidae								
<i>Urocissa caerulea</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[148]
<i>Gymnorhinus cyanocephalus</i>	D	—	R	F	K	pair nesting with related male helpers	1	[149, 150]
<i>Aphelocoma californica</i>	D	—	R	F	K	pair nesting with (presumed) related male and female helpers	1	[151]
<i>Aphelocoma coerulescens</i>	D	—	G	F	K	pair nesting with related male and female helpers	1	[152, 153]
<i>Aphelocoma ultramarina</i>	D, I	50%	G	O	M	pair nesting with related and unrelated male and female helpers; polyandrous mating	2	[154, 155]
<i>Cyanocorax beecheii</i>	D	—	R	F	K	pair nesting with related male and female helpers	1	[156, 157]
<i>Cyanocorax yncas</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[158, 159]
<i>Psilorhinus morio</i>	D, I	40%	G	O	M	cooperative polyandry with related female and unrelated male helpers	4	[160-162]
<i>Calocitta formosa</i>	D, I	frequent ¹²	G	O	M	pair nesting with related and unrelated female helpers; polyandrous mating and occasional joint-nesting; conspecific brood parasitism	2	[163-165]
<i>Perisoreus canadensis</i>	D, I	25%	R	F	M	pair nesting with related and unrelated helpers	2	[166]
<i>Cyanopica cyana</i>	D, I	25-75%	R	F	M	pair nesting with related and unrelated male and female helpers	2	[167, 168]
<i>Corvus caurinus</i>	D	—	R	F	K	pair nesting with related helpers	1	[169]
<i>Corvus brachyrhynchos</i>	D	—	G	F	K	pair nesting with related male helpers	1	[170]
<i>Corvus corone</i>	D, I	50%	G	F	M	cooperative polyandry by unrelated males; related male helpers	4	[171, 172]

¹⁰ Kidnapping of unrelated young, and the formation of polygamous coalitions by non-relatives, are uncommon but regular occurrences [139].

¹¹ Percentages not given, but turnover among adult group members was very common, with multiple transitions between groups observed across years. Same-sex group members sometimes joined together, suggesting possible sibling dispersal [146].

¹² Relatedness of helpers to the young that they provision is low, “averaging roughly half that of first cousins” [165].

Corvoidea incertae sedis								
<i>Mohoua albicilla</i>	D	—	R	F	K	pair nesting with related male helpers	1	[173]
Pithacartidae								
<i>Chaetops frenatus</i>	D	—	B	F	K	pair nesting with related male and female helpers	1	[174]
Petroicidae								
<i>Melanodryas cucullata</i>	D	—	B	F	K	pair nesting with (presumed) related male helpers	1	[175]
<i>Melanodryas vittata</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[176]
<i>Eopsaltria australis</i>	D	—	R	F	K	pair nesting with related male helpers	1	[177]
<i>Eopsaltria georgiana</i>	D	—	R	F	K	pair nesting with related male helpers	1	[178]
Paridae								
<i>Pseudopodoces humilis</i>	D, C, I	frequent ¹³	G	F	M	cooperative polyandry with related and unrelated males	4	[179-181]
<i>Parus niger</i>	D	frequent ¹⁴	R	F	M	pair nesting with related and unrelated male helpers	2	[182]
<i>Parus fasciiventer</i>	D	—	R	F	K	pair nesting with related helpers	1	[183]
<i>Anthoscopus minutus</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[184]
Aegithalidae								
<i>Aegithalos caudatus</i>	D	—	G	F	K	pair nesting with related male and female helpers	1	[185, 186]
<i>Psaltriparus minimus</i>	D, I	frequent	R	F	M	pair nesting with unrelated male and female helpers	2	[187, 188]
Pycnonotidae								
<i>Alophoixus pallidus</i>	D	—	R	F	K	pair nesting with related male helpers	1	[189]
<i>Phyllastrephus cabanisi</i>	D	—	G	F	K	pair nesting with related male helpers	1	[190]
Cisticolidae								
<i>Cisticola chinianus</i>	D, C	43%	R	F	M	cooperative territory defense by related or unrelated pairs of males	4	[191]
Timaliidae								
<i>Yuhina brunneiceps</i>	C, I	majority	G	O	NK	joint nesting by unrelated pairs	3	[192, 193]
<i>Turdoides caudatus</i>	D, I	33%	R	F	M	pair nesting with related male and unrelated female helpers; cooperative polygyny by unrelated females	4	[194]

¹³ Helpers are males whose broods have failed; related helpers are more likely to gain paternity in the brood than are unrelated males.

¹⁴ Helpers are retained male offspring, but Tarboton [181] concluded that relatedness between helpers and the brood should be low due to high mortality and turnover of adult group members.

<i>Turdoides squamiceps</i>	D, I	20-35%	G	O	M	pair nesting with related helpers; cooperative polyandry by related and unrelated males	4	[195, 196]
<i>Turdoides fulvus</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[197]
<i>Turdoides striatus</i>	D	—	R	F	K	pair nesting with related helpers	1	[198]
<i>Turdoides bicolor</i>	D	—	G	O	K	pair nesting with related male and female helpers	1	[199, 200]
<i>Turdoides gymnogenys</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[201]
<i>Babax waddelli</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[202]
Acrocephalidae								
<i>Acrocephalus melanopogon</i>	C	majority	G	F	NK	cooperative polyandry by unrelated males	3	[203]
<i>Acrocephalus vaughani taiti</i>	C	majority	G	F	NK	cooperative polyandry by unrelated males	3	[204]
<i>Acrocephalus baeticatus</i>	C	majority	G	F	NK	cooperative polyandry by unrelated males	3	[205]
<i>Acrocephalus sechellensis</i>	D, C, I	frequent	G	F	M	pair nesting and cooperative polygyny with related and unrelated male and female helpers; polygynandrous mating	4	[206]
Passerida incertae sedis								
<i>Erythrocercus mccallii</i>	D	—	B	F	K	pair nesting with related helpers	1	[207]
<i>Donacobius atricapillus</i>	D	—	R	F	K	pair nesting with related helpers	1	[208]
Troglodytidae								
<i>Campylorhynchus griseus</i>	D	—	G	F	K	pair nesting with related male and female helpers	1	[209]
<i>Campylorhynchus nuchalis</i>	D, C, I	7.5-30%	G	F	M	pair nesting with related male and female helpers; cooperative polyandry by unrelated males	4	[210-212]
<i>Campylorhynchus fasciatus</i>	D	—	R	F	K	pair nesting with related male and female helpers	1	[213]
<i>Campylorhynchus yucatanicus</i>	D	—	R	F	K	pair nesting with related male helpers	1	[214]
<i>Campylorhynchus zonatus</i>	D, I	frequent ¹⁵	G	F	M	pair nesting with related and unrelated male and female helpers; conspecific brood parasitism	2	[215]
Sittidae								
<i>Sitta pygmaea</i>	D	—	R	F	K	pair nesting with related male helpers	1	[216, 217]

¹⁵ Based on small sample size. Skutch (cited in [215]) observed three nesting pairs with helpers. At two nests he assumed that the helpers were retained offspring of the pair, but at the third nest the helper joined the pair very late in the nesting season and Skutch concluded that it was an unrelated failed breeder. More recently, M. Wells (pers. comm.) genetically sampled four family groups and found that 2 of 8 nestlings were unrelated to any adult group member, suggesting conspecific brood parasitism.

<i>Sitta pusilla</i>	D, I	38%	G	F	M	pair nesting with related and unrelated male helpers	2	[218, 219]
Sturnidae								
<i>Lamprotornis nitens</i>	D	—	B	F	K	pair nesting with related helpers	1	[220]
<i>Lamprotornis superbus</i>	D, I	30-70% ¹⁶	G	O	M	pair nesting with related and unrelated male and female helpers; cooperative polyandry by unrelated males	2	[221, 222]
<i>Lamprotornis pulcher</i>	D, I	frequent ¹⁶	R	O	M	pair nesting with related and unrelated male and female helpers	2	[223]
<i>Spreo bicolor</i>	D, I	frequent ¹⁶	R	O	M	pair nesting with related and unrelated male and female helpers	2	[224]
<i>Cosmopsarus regius</i>	D	—	B	O	K	pair nesting with (presumed) related helpers	1	[225]
<i>Cosmopsarus unicolor</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[226]
<i>Buphagus africanus</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[227]
Mimidae								
<i>Mimus gilvus</i>	D	—	G	F	M	pair nesting or cooperative polygynandry by unrelated adults with related male and female helpers	4	[228]
<i>Nesomimus parvulus</i>	D, I	27-42%	R	F	M	pair nesting or cooperative polygynandry with related and unrelated male and female helpers; plural nesting	4	[229]
<i>Nesomimus trifasciatus</i>	D	—	B	F	K	pair nesting with (presumed) related male helpers	1	[230]
<i>Nesomimus macdonaldi</i>	D, C, I	majority	G	F	M	cooperative polygynandry with related and unrelated male and female helpers; plural nesting; conspecific brood parasitism	4	[231]
<i>Ramphocinclus brachyurus</i>	D	—	G	F	K	pair nesting with related male and female helpers	1	[232]
Muscicapidae								
<i>Erythropygia coryphaeus</i>	D	—	G	F	K	pair nesting with related male helpers	1	[233, 234]
<i>Oenanthe lugubris</i>	D	—	B	F	K	pair nesting with related male helpers	1	[235]
<i>Myrmecocichla formicivora</i>	D	—	G	F	K	pair nesting with related male helpers	1	[236]
Turdidae								
<i>Catharus bicknelli</i>	C	majority	G	F	NK	cooperative polyandry by unrelated males; polygynandrous mating	3	[237]
<i>Sialia mexicana</i>	D	—	G	F	K	pair nesting with related male helpers	1	[238-240]
Nectariniidae								
<i>Nectarinia dussumieri</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[241]
Prunellidae								

¹⁶ Helpers attend many nests simultaneously; male helpers are often related to the broods they help, whereas female helpers are less common and typically unrelated to the broods that they helped. Turnover in group membership implies that male helpers probably assist at many nests with varying degrees of genetic relatedness.

<i>Prunella collaris</i>	C	majority	G	F	NK	cooperative polyandry by unrelated males	3	[242-244]
<i>Prunella fulvescens</i>	C	majority	R	F	NK	cooperative polyandry by unrelated males	3	[245]
<i>Prunella modularis</i>	C	majority	G	F	NK	cooperative polyandry by unrelated males	3	[246]
<i>Prunella rubida</i>	C	majority	B	F	NK	cooperative polyandry by (presumed) unrelated males	3	[245]
Ploceidae								
<i>Bubalornis niger</i>	C	90%	G	O	NK	cooperative polyandry by unrelated males; polygynandrous mating	3	[247]
<i>Plocepasser mahali</i>	D	—	R	F	K	pair nesting with related male and female helpers	1	[248, 249]
<i>Pseudonigrita arnaudi</i>	D	—	R	F	K	pair nesting with related male helpers	1	[250, 251]
<i>Philetairus socius</i>	D	—	G	F	K	pair nesting with related male and female helpers	1	[252, 253]
Fringillidae								
<i>Loxioides bailleui</i>	D	—	G	F	K	pair nesting with related male helper	1	[254]
<i>Paroreomyza montana</i>	D	—	G	F	K	pair nesting with related male and female helpers	1	[255]
Emberizidae								
<i>Calcarius pictus</i>	C	majority	G	O	NK	cooperative polyandry by unrelated males; polygynandrous mating	3	[256, 257]
<i>Ammodramus savannarum</i>	I	majority	R	F	NK	pair nesting with unrelated helpers	2	[258]
<i>Neothraupis fasciata</i>	D	—	B	F	K	pair nesting with (presumed) related male and female helpers	1	[259, 260]
<i>Cypsnagra hirundinacea</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[261]
<i>Tangara cyanoventris</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[261]
<i>Tangara larvata</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[261]
<i>Geospiza scandens</i>	I	majority	R	F	NK	pair nesting with unrelated male helpers	2	[262]
<i>Caryothraustes poliogaster</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[263]
Icteridae								
<i>Hypopyrrhus pyrohypogaster</i>	D	—	R	O	K	pair nesting with related male and female helpers	1	[264]
<i>Pseudoleistes virescens</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[265]
<i>Pseudoleistes guirahuro</i>	D	—	B	F	K	pair nesting with (presumed) related helpers	1	[265]
<i>Agelaioides badius</i>	D, I	25%	R	F	M	pair nesting with related and unrelated male and female helpers; cooperative polyandry and joint nesting rare	2	[266]

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